

EKL Safety Training

Safety is the responsibility of everyone, everywhere.



January 2023 – Cryo Lab version

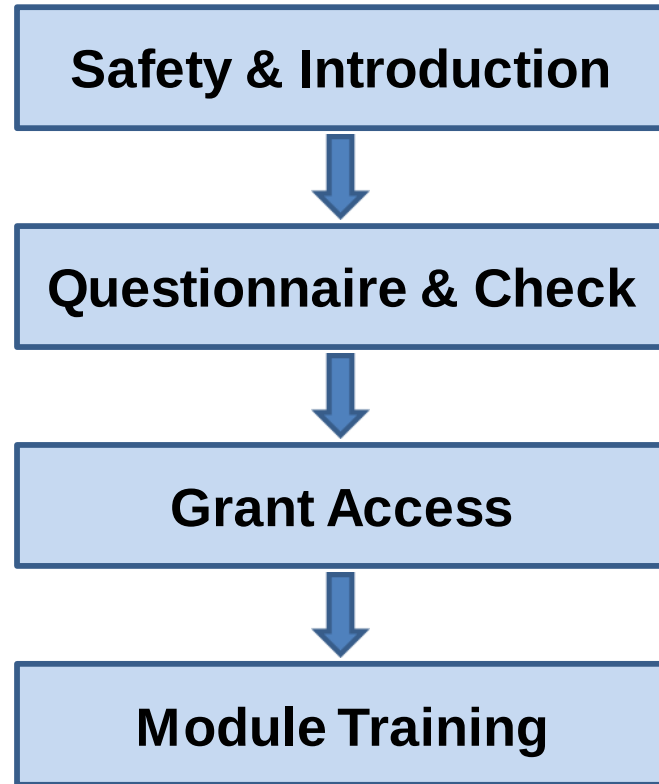
Introduction

During this course you will be informed about the safety rules and precautions in the EKL cleanrooms. You must learn to work safely.

- The safety of the lab and its users are the most important.
- You must always think about safety with every action that you take.



Introduction

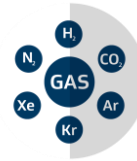


Additional training must be followed to use any lab equipment according to its manual, to prevent unsafe situations or damage to the systems.

Safety risks

Working in a cleanroom or laboratory results in a higher safety risk than working in an office environment:

- **Liquids** and **solids** are being used.
- **Gases** are being used.



Some of these are **hazards** !!!

EKL safety measures

EKL has the following safety measures:

- Legislation from the Government and the City of Delft.
- An Access Control System.
- An Alarm System with Alarm Buttons and a Gas Detection system.
- An Emergency Response Team (in Dutch: BHV = Bedrijfs Hulp Verlening).
- Education and training of users.
- A Safety Officer and a Safety Committee.

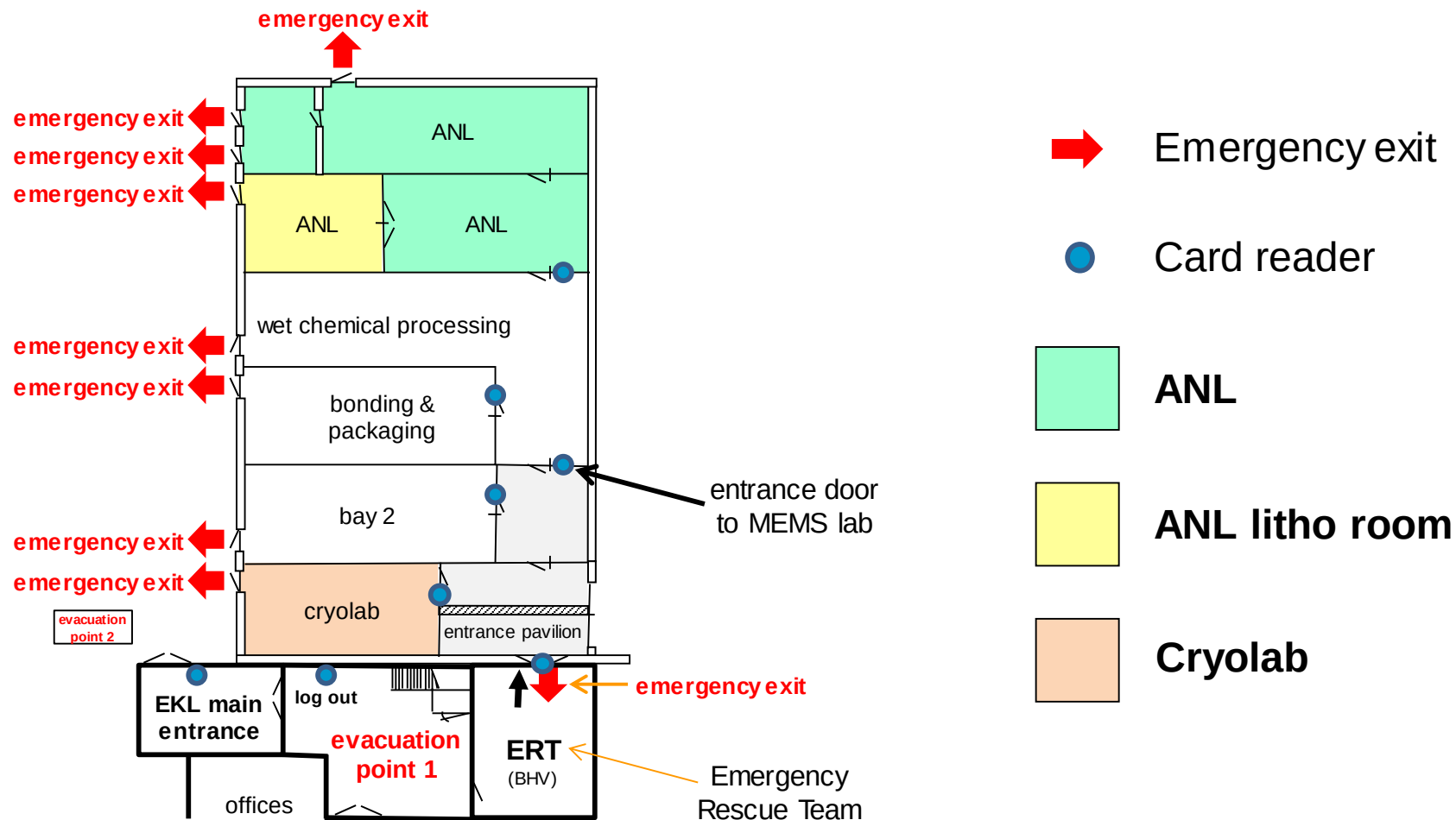


Requirements to work in the labs

- Become familiar with the **map of the labs**.
- **Know where to find:**
 - emergency exits
 - alarm buttons
 - fire extinguishers
 - emergency kits
 - showers
- **Keep access free to:**
 - escape routes
 - fire extinguishers
 - emergency kits
- Always follow the instructions of the **Emergency Response Team (ERT)**.



Pavilion/MEMS escape routes in case of an alarm



Working with chemicals

Rules

- Everyone needs a **buddy** to work with chemicals or liquid Nitrogen.
- Wear **safety glasses** or a **protective face shield** (also when you are looking over someone's shoulder).
- Wear **neoprene gloves**.
- Wear protective clothing (**Apron**).
- Wear trousers and closed shoes. (**Shorts and open shoes are not allowed**).

Your clothes are the only absorbent in case of a chemical or liquid nitrogen spill.



Emergency Response Team

The ERT consists of trained staff members who will help with incidents ranging from personal injuries and chemical incidents, to extinguishing a fire and rescuing people.

ERT members have this sign on their office



Access control

The access control system:

The main purpose of access control is to **provide security** to the EKL users and visitors by allowing or restricting access to the labs.

The following rules apply:

- Each lab user must use a **valid badge** to enter or exit a lab.
- It is **not allowed** that 2 or more users enter a lab with one badge.
- Visitors & guests must be escorted by a staff member or **qualified** user.

When entering or leaving a lab:

- **Sign in:** Use your **own** access badge to enter the room.
- **Log out:** Use your **own** access badge when leaving the room.

In case of an alarm or incident the Emergency Response Team and/or the Fire Brigade can check who is still where in the EKL building.



Type of alarms

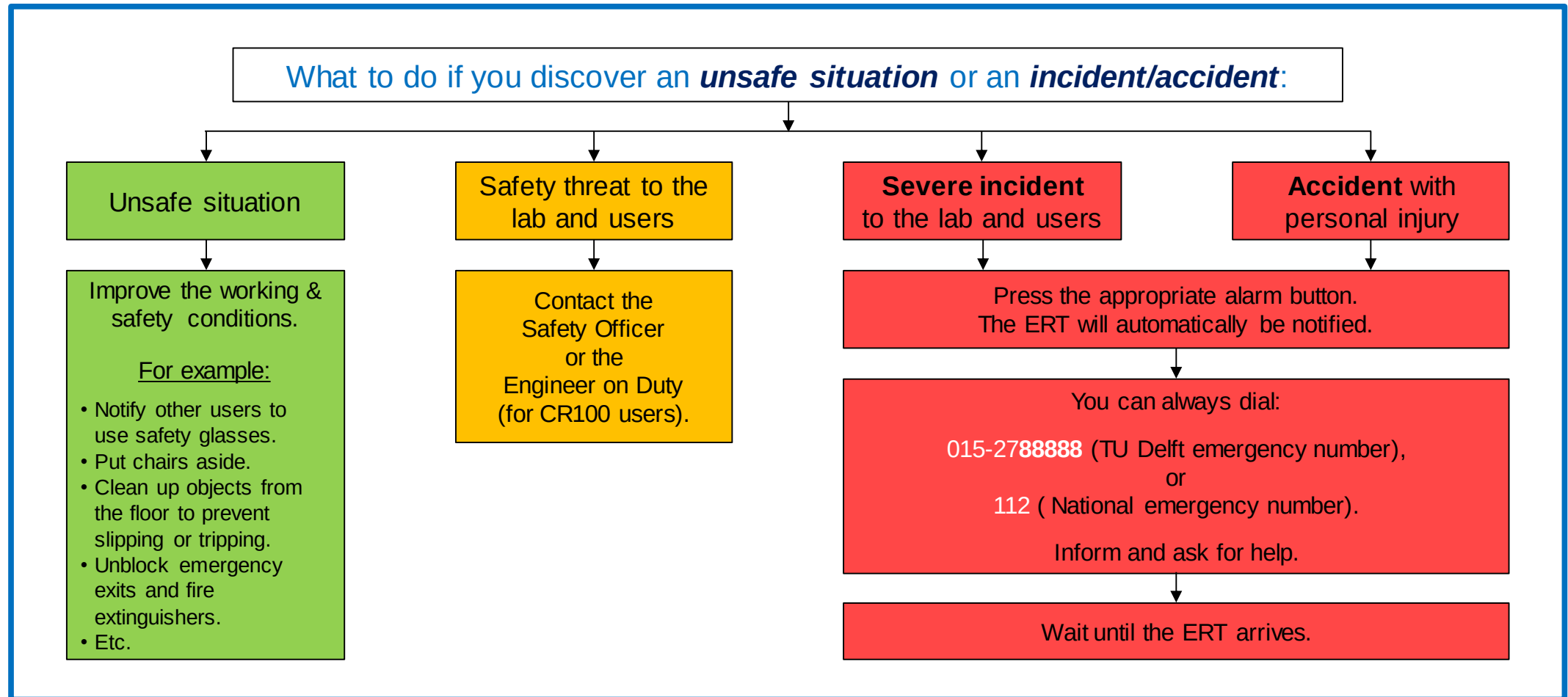
- Automatic smoke detector alarm.
This activates an alarm sound, and on the alarm panels only the text "Paviljoen verlaten" (= leave pavilion) will light up.
- Automatic gas detector alarm.
This activates an alarm sound, and on the alarm panels both the text "Gasalarm" and "Paviljoen verlaten" will light up.
- Manual alarm (pressed button).
This activates an alarm sound, and on the alarm panels only the text "Paviljoen verlaten" will light up.



During alarm: Sound + Panel illuminated.

An alarm test is performed every first Monday of the month at 8:30 hrs.

Overview of safety procedures



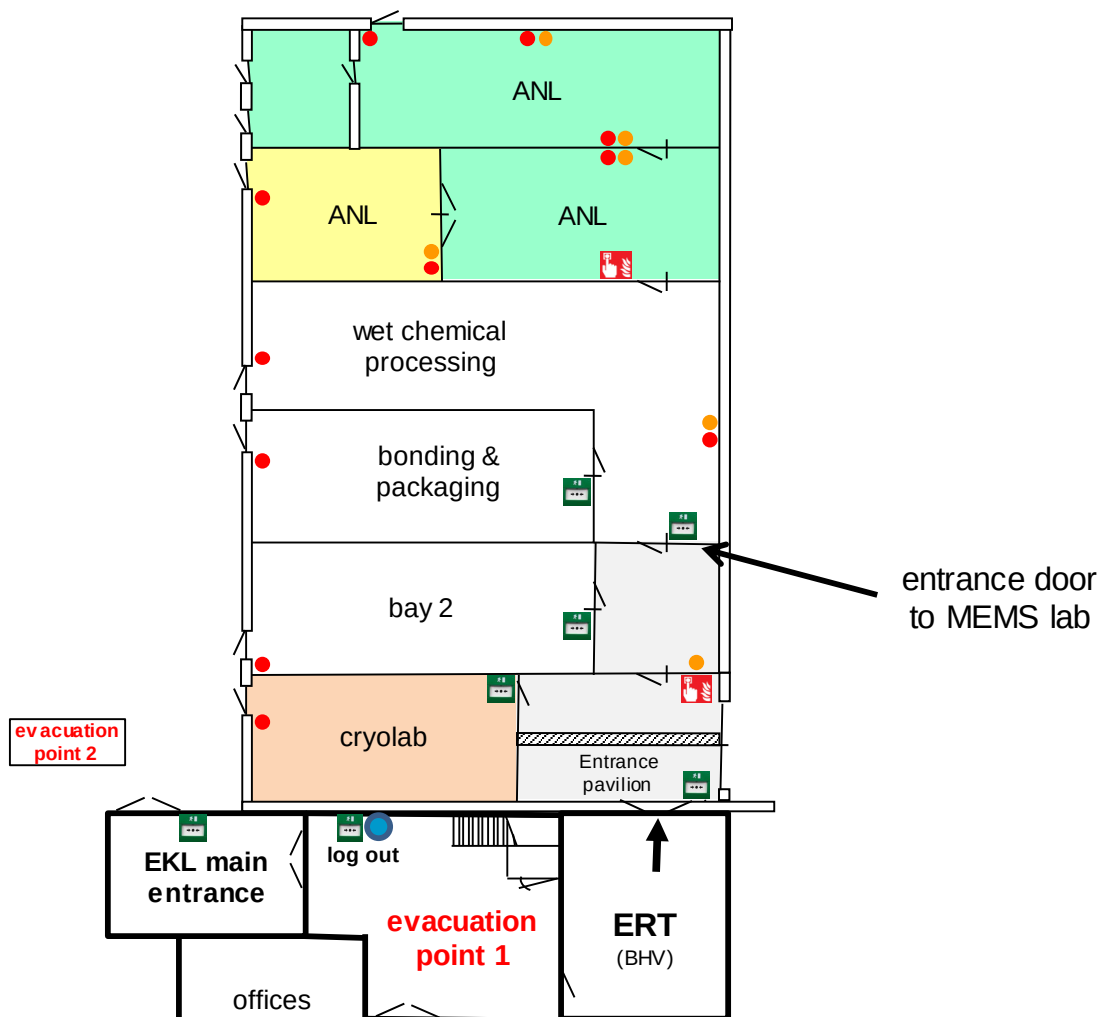
All safety incidents and accidents must be reported on SharePoint.

Function of the alarm buttons in EKL

Name	Picture	When	Result	Additional
Calamity button		If you need immediate help from the ERT (e.g. accident with a person, chemical spill).	- An alarm sounds and the sign "Paviljoen verlaten" lights up. - The ERT is alarmed. - The lab must be evacuated. - All standard access doors will be unlocked.	If somebody is injured then provide help until the ERT arrives, and think of your own safety.
Emergency button		In case of a gas leak or other severe threat to the lab and its users (e.g. large chemical spill generating dangerous fumes).	- An alarm sounds and the sign "Paviljoen verlaten" lights up. - The ERT is alarmed. - All labs and basement must be evacuated. - All standard access doors will be unlocked. - All gas valves are closed (except N2). - Air recirculation stops: Only fresh air goes in.	If somebody is injured then provide help until the ERT arrives, and think of your own safety.
Fire Alarm button		In case of fire.	- An alarm sounds and the sign "Paviljoen verlaten" lights up. - The ERT is alarmed. - All labs and basement must be evacuated. - All standard access doors will be unlocked. - All gas valves are closed (except N2). - Air recirculation stops: Only fresh air goes in. The Fire Department is alarmed.	If safe and possible (for waste basket size fires), extinguish the fire with a fire extinguisher. If extinguishing is no longer possible or safe then stop and leave room.
EMO button		In case of a dangerous situation with a machine (e.g. smoke comes out, a person is trapped inside system, liquid is leaking).	No cleanroom alarm is generated. - Only for this tool: System will switch off (i.e. no electricity is present in the system after the main power switch).	Warn the tool owner.

Important: Inform the ERT at the evacuation point why you pressed the alarm.

Location of the alarm buttons



Calamity button



Emergency Stop button



Fire Alarm button
(usually next to an exit door)



Door unlock button
(if not automatically unlocked)



Be aware where the alarm buttons are located!

What to do if there is an alarm

- Leave the work area immediately. Do not finish your process.
- Alert other users on your way out.
- Follow the escape route to the nearest emergency exit.
 - All doors will automatically unlock.
(If not, then press the button in the green box next to the door.)
 - Close the doors behind you.
 - Do **NOT** take off your cleanroom suit or lab coat.
- Go to evacuation point 1 and log out.
 - Here users can take off their cleanroom suit or lab coat.
- Wait for further instructions.

Note: Evacuation point 2 is only used if point 1 is unsafe.

If you know what caused the alarm then inform the ERT.



Sign indicating the escape route



Green box to unlock the door
(if not automatically unlocked)



Evacuation point 1

inside the building at
the reception hall



Evacuation point 2

outside the building in front
of the main entrance

How to handle in case of an incident/accident **outside office hours**

1. Press the appropriate emergency button.
2. In case of a *severe incident to the lab and users* or in case of an *accident with personal injury*:
 - Dial: **015-2788888** TU Delft Emergency number.
 - Dial: **112** National Emergency number.
3. Supply the following information to the operator:
 - Your name.
 - The address of EKL ([Feldmannweg 17, 2628CT Delft](#)).
 - Location of the incident (Office, CR100, CR10000, MEMS, basement, etc.).
 - The nature of the incident (e.g. hazardous materials which are involved).
 - If an ambulance is needed (to your judgement).
4. Do **NOT** go home:
 - Wait for the ERT to arrive, this can take up to 20 minutes.
 - Supply them with detailed information about the incident or accident.

Outside office hours are: Weekdays from 18.00 - 8.00, weekends and holidays.

What to do in case of a chemical spill outside a fume hood or a wet-bench

How to react:

- Press the **CALAMITY** button.
- Think about your own safety and avoid breathing in vapors from the spill.
- Evacuate the laboratory and warn people to avoid that someone gets contaminated or slips over the spilled chemical.
- Follow the [What to do if there is an alarm](#) procedure. (See slide 15 or click on the link.)
- Inform the ERT at the evacuation point.
- Report the incident on SharePoint.

Chemical spill on a person

Reaction as a victim:

- Press the **CALAMITY** button.
- Try not to panic.
- Ask another user for help (buddy).
- Move away from the splashed chemical, if possible into a safe area.

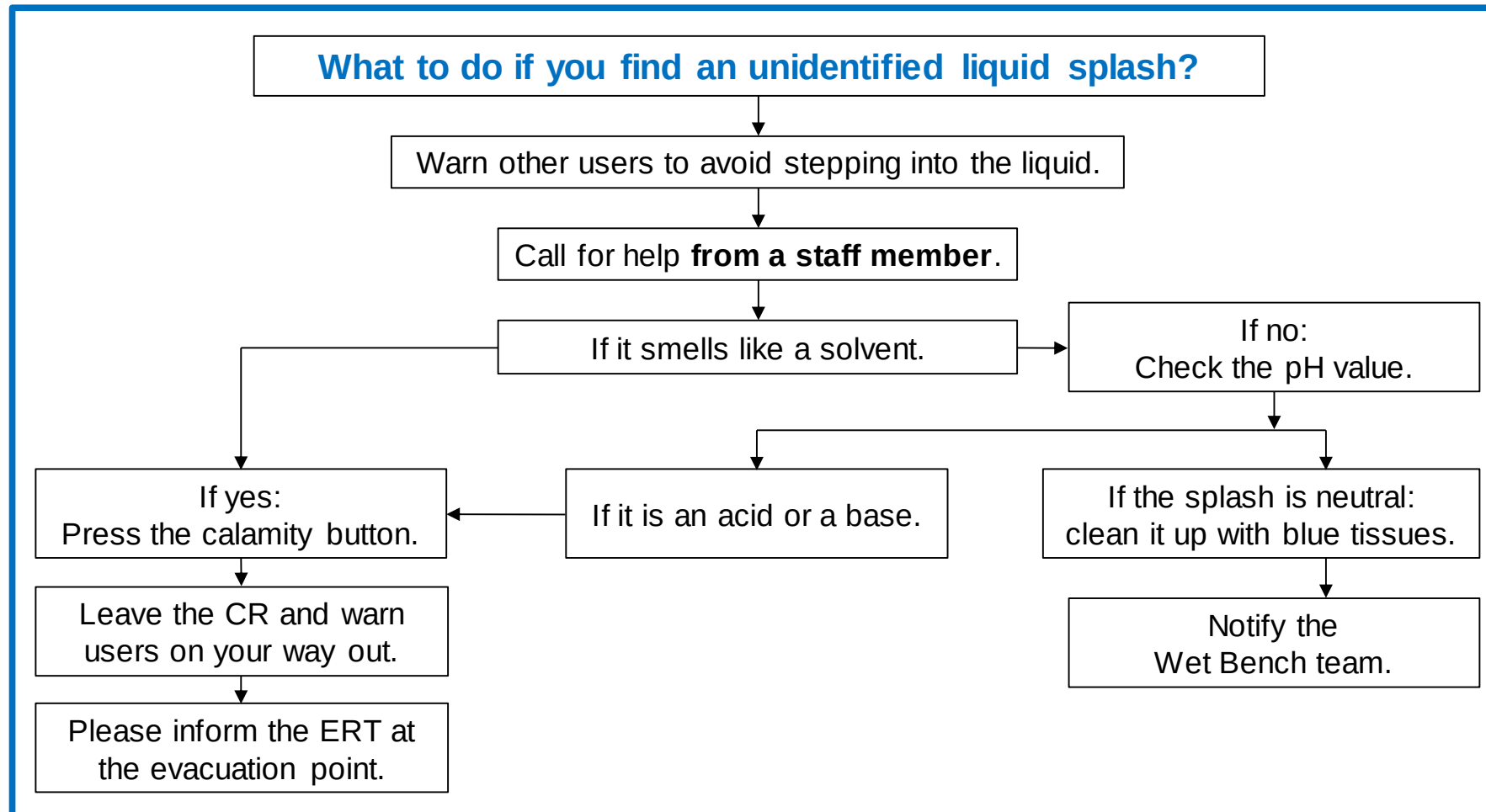
Reaction as a helper:

- Press the **CALAMITY** button.
- Think about your own safety.
- Stay with the victim and try to help.
- Ask another user for help (buddy).
- Use neoprene gloves from the emergency kit.
- Move the victim away from the splashed chemical, if possible into a safe area.

In both cases:

- Apply the appropriate neutralizing solution on the victim (on clothes if necessary):
 - Organic contamination (e.g. for acetone, NMP, IPA): → Use Diphoterine or Water.
 - Acid or a Base contamination (e.g. nitric acid, KOH, TMAH): → Use Diphoterine.
 - HF contamination (e.g. Poly-Si etch, BHF, HF): → Use Hexafluorine.
 - If you get a drop in your eye: → Use the eye bottle to flush. Use the whole bottle, but do not squeeze.
- Remove the victims clothing, including boots & shoes (use scissors if necessary ➡ available in rescue kit). Try to prevent further spreading out of the chemical over the victim.
- Wait for help from the ERT.
- Report the accident on SharePoint.

What to do if you find an unidentified liquid splash?



All safety incidents and accidents must be reported on SharePoint.

Reporting of incidents or accidents, and improvements & suggestions

Create a report using the template on SharePoint:

<https://sharepoint.tudelft.nl/sites/EKL2/Safety/SitePages/Home.aspx>

Click the button to report an incident or accident!



- All incidents or accidents **must** be reported. They will be analyzed and an action list will be defined.
- You can follow the status of your reported incident on:

<https://sharepoint.tudelft.nl/sites/EKL2/Safety/IncidentsFiles/Forms/MyIncidents.aspx>

Liquid Nitrogen (N₂) spill

Spillage or leakage of liquid N₂ can result in a lack of sufficient oxygen and cause respiratory problems.

1 liter of liquid N₂ produces **700 liters** of N₂ gas.

A room where liquid N₂ is being used has an O₂ detector that triggers an alarm if the Oxygen level is lower than 19%.

- If you see an unconscious person in a room with low oxygen content: **Do not enter.**
- Press the calamity button.
- Leave the area and alert people on your way out.
- At the evacuation point inform the ERT why you pressed the calamity button.
- Report the incident or accident on SharePoint (according to the protocol).

It is not allowed to use liquid N₂ without permission and proper training.



Liquid Nitrogen vessel



O₂ detector



O₂ detector in basement



N₂ tap point

